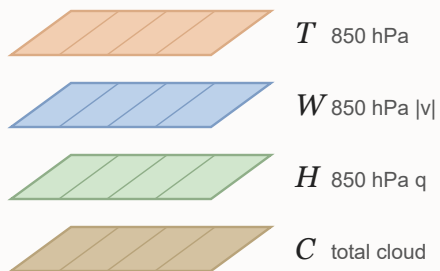


## Regional data



ERA5 1979–2025 · 6-hourly ·  $64 \times 32$

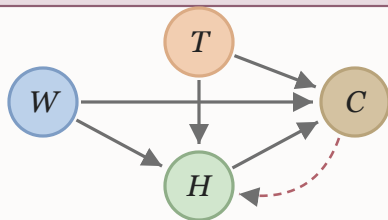


**66 regions × 4 variables**

anomalies fitted on 1979–2018 only

+  $u$ ,  $v$ ,  $qu$ ,  $qv$  · six ERA5 controls  
 + CERES satellite cloud 2017–2025

## Stage 1 · Screen



physics-guided candidates,  $k = 2$  neighbours  
 dashed: reverse candidates (direction test)

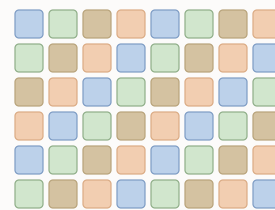
$$y_t = \alpha + \gamma^T c_t + \beta x_{t-\ell}$$

$c$  = target's own lags 1–3 (recall)  
 calendar-time HAC · BH  $q < 0.05$

**2,187 / 2,574**

edge-lag hypotheses, 1979–2018

## Stage 2 · Confirm



dense VAR(3) on all 264 regional histories

$$X_t = \sum_{\ell \leq 3} A_\ell X_{t-\ell} + \varepsilon_t$$

removes indirect and shared-history edges  
 HAC · BH  $q < 0.05$  (precision)

**1,550 confirmed**

screened and confirmed, 1979–2018

## Stage 3 · Replicate

1979–2018 · fit

2019–2023 · held out, primary  
 2023–2025 · held out, second

frozen preprocessing and selection

same sign + one-sided HAC test,  
 BH within the confirmed family

frozen-coefficient predictive gain  
 unconfirmed hypotheses = negative control

**760 / 1,550 replicate**

control: 95 / 1,024 (2019–2023)

## Validation suite around every stage



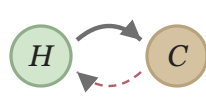
### Calibration

known-null FDR  
 known-structure F1



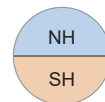
### Specificity

100 type- and distance-  
 matched candidate graphs



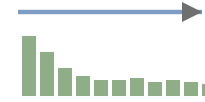
### Direction

$W \leftrightarrow H$ ,  $H \leftrightarrow C$ ,  $W \leftrightarrow C$  under  
 full conditioning



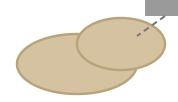
### Regimes

state interactions, 24 h prior  
 and Niño3.4 states, hemispheres



### Timing & transport

lag windows, aligned paths,  
 vector wind, moisture flux



### Independent cloud

CERES substitution,  
 same-date source check